DRAFT (deadend) — Bimodal Semantic Labor Measure

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Abstract

This negative-result draft tests whether task directionality can be measured without inferring a task vector. It represents user input and model output as two semantic-labor vectors and initially proposes cosine similarity between them. The attraction is methodological: if both sides are directly measured, the contaminated intermediary "what was the task?" might disappear.

The document then runs its required Irish-famine reproduction gate and records failure. The base case scores 0.928 rather than the target 0.80; the contrast case scores 0.739 rather than 0.25. Neutral offers inherit topical similarity and become falsely positive, while operational opposition shares topic features and is insufficiently negative. Raw cosine therefore cannot distinguish neutral from advancing or topical relevance from opposition.

The draft concludes that recovering the needed sign requires an intent baseline, restoring the very task direction the symmetric formulation tried to eliminate. The surviving insight is transferred to the Kuro bridge: input can supply the baseline axis, but the pure two-vector cosine measure is decorative and should not proceed. This is a valuable failed gate and should remain cataloged as a negative result.

Body

DRAFT — Bimodal Semantic Labor Measure

Status: working draft, not deposited, separate from the Directionality of Semantic Labor spec (DOI 10.5281/zenodo.20469514). Provisional name; conjectural. Requires the famine-case reproduction test (below) before any deposit consideration.

Author register: TBD (Lee Sharks aperture / measurement domain) — not assigned.

The move

DSL as deposited routes through an intermediary: it scores output against a *task vector* T, which on unprimed dialogue must be *inferred*. Every identification problem the program has hit — the ΔG shortcut, the $A<_n$ inference contamination, frame endogeneity in reflexive threads, the laundering risk in retrocausal stabilization — is a **task-inference** problem. T is the soft joint where contamination enters.

This draft asks: what if the metric does not route through T at all?

Frame the measured quantity as **semantic labor** directly, and compute **both sides** of the exchange as labor. The user's input is labor — it has a direction and magnitude in semantic space. The model's output is labor — same. Then measure the *relationship between the two labor vectors*, not the output against an inferred abstraction of the input. ## Why this dissolves rather than bounds the loop

When both sides are the same kind of measured object, there is no inferred intermediary to contaminate. T disappears as a separate entity. Directionality becomes the relationship between two *observable* vectors, both read off the transcript, neither the contested quantity — extending to the whole metric the property that made Lead-Lag Drift Attribution identified (its inputs, turn embeddings and order, are not disputed).

Calculating the **input** labor is the load-bearing half, not a symmetric nicety. At present the input is treated only as a *source of T*: a task vector is squeezed out and the rest discarded. That squeezing **is** the inference step, and the inference step is where the leaks live. Computing the input as labor in its own right — its own directional vector L_in — makes the input one of two measured quantities rather than raw material for inference. T ceases to be a separate object. ## Sketch (conjectural — geometry to be tested, not asserted)

For an exchange, let L_in be the input labor vector and L_out the output labor vector, both under a frozen, declared representation δ (same discipline as DSL's δ; see risk 1).

- **Advancing labor:** L_out extends L_in in its own direction (positive projection).
- **Redirecting labor:** L_out orthogonal to or opposed to L_in.
- Directionality ≈ a function of the projection / angle of L_out onto L_in (exact geometry TBD).

This is symmetric and coder-independent in the way Lead-Lag is: both vectors are computed, neither inferred. ## What it does to the retrocausal problem

There is nothing to stabilize retroactively because there is no latent task awaiting clarification. Input labor is fully present in the input — a computed quantity, not a hidden vector inferred-then-corrected. A user's later correction is not "retroactively clarifying what the task was"; it is simply *more input labor* — a new L_in, measured the same way. "Discovery through friction" is not retrocausation; it is the input-labor vector changing direction across turns, measured directly. ## Risks (the reasons this might be decorative rather than real)

- **Representation smuggling.** Computing L_in/L_out as vectors requires an embedding/δ. If that representation is model-produced and free, the contamination returns one level down. Defense: the frozen-δ discipline already in DSL — declared, version-pinned, reproducible. Consequence: the "no inference" claim is precisely "no *task* inference"; a representation step remains and must be the declared, auditable kind. The orthogonality condition from the DSL kernel does not vanish — it relocates to "the labor representation must be frozen and external," which is a cleaner home for it.
- **Magnitude opens a second axis.** Labor has magnitude as well as direction. Measuring input labor invites measuring whether output labor was *proportionate* to input labor — adjacent to but not identical with directionality. This is where ULD and the work-rate operators (WRS/PVS) independently arrived; they are labor-*magnitude* operators. So

input-labor-as-measured-quantity may be the common substrate beneath DSL (direction-relationship), ULD (input labor diverted to substrate-management), and WRS (output labor rate vs input labor commissioned). **Flagged as a unification temptation and distrusted accordingly** — fundamental objects do legitimately unify, which is exactly why a seductive unification must be tested, not adopted.

The clean test (gate before any deposit)

Reproduce the known case. The deposited DSL worked example (Appendix D, Irish famine, one-shot) scores DSL +0.80 by span classification. Construct L_in from the commission and L_out from the five-span response under a declared δ, compute the projection geometry, and check whether it reproduces ≈ +0.80 and the contrast-case drop to +0.25.

- If it reproduces the known scores: the vector framing holds numbers and may be real.
- If it cannot: "labor as a vector" is a metaphor that won't compute, and the framing is decorative.

No deposit until the known case reproduces.

RESULT — failed gate, recorded (2026-05-31)

The famine reproduction test was run. **The bimodal measure as drafted does not reproduce the known case, and the failure has a specific, informative cause.**

Setup: L_in built from input demand-features only (operation=enumerate, object=causes-of-famine, concision, on-topic), L_out per span on the same axes, directionality = cosine(L_out, L_in). No task vector supplied — the pure-bimodal condition.

Results: base case **+0.928** against the +0.80 target (overshoot); contrast case **+0.739** against the +0.25 target (worse overshoot); base→contrast drop only 0.19 vs DSL's 0.55. The measure is systematically too generous and under-discriminating.

Two failure modes, both diagnostic:

- **Neutral inflates.** "Happy to go deeper" scores +0.66, not 0. Cosine cannot represent neutral-as-zero unless the span is engineered exactly orthogonal, and natural offers are weakly aligned with the task, not perpendicular. Cosine turns neutral labor into mild-positive labor.
- **Opposition dilutes.** An oppositional span scores ≈ −0.4, not −1, because it still shares topical axes (on-famine, concise) with L_in even while opposing the *operation*. Cosine over a shared feature basis cannot separate "opposes the task" from "is about the same subject"; topical overlap dilutes opposition.

Diagnosis. Cosine similarity is the wrong relationship operator. The two distinctions DSL's discrete taxonomy is built to make — neutral-as-zero and opposition-as-signed-negative-independent-of-topic — are exactly what raw vector similarity loses. The fix is a *signed*

projection onto a directional baseline that distinguishes orthogonal from aligned and penalizes operational opposition independent of topical overlap. But projection onto a baseline **requires a baseline direction — i.e. intent**.

Consequence for the bimodal program. The bimodal aim was to eliminate the task vector entirely and measure only the relationship between two input/output-derived labor vectors. The famine test shows that removing the task vector loses the neutral/oppositional discrimination, and restoring that discrimination reintroduces intent as the projection axis. The bimodal measure therefore does not stand as a symmetric two-vector operator; it **reduces to the intent-baselined projection** — which is the Kuro-bridge formulation (project output labor onto the commission baseline), and which *did* reproduce +0.80 on-axis.

Verdict. Decorative in the symmetric (no-baseline, cosine) form. The surviving form of the idea is the intent-baselined signed projection — already instantiated in the Kuro bridge and shown there to be the taxonomy-quantized member of the deviation family. The pure-bimodal symmetric measure is recorded here as a dead-end with cause, not carried forward. The salvage: input *can* be computed as labor (L_in is constructible), but it functions as the baseline axis, not as a co-equal second vector under similarity.

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